

Occurrence of Diopside and Cr-Zr-Armalcolite on the Moon

SEVERAL suggestions have been made, on the basis of petrological evidence (for example, refs. 1-3), to the effect that the Moon underwent extensive differentiation early in its history. The early products of such crystal-liquid differentiation should contain ferromagnesian minerals with a high Mg/Fe value and plagioclases with a high An content. They would occur at depth, by analogy with the terrestrial mantle, and should therefore be uncommon in the Apollo samples. Petrographic and electron-microprobe examination of the sample of 1 to 2 mm fines, Apollo 14166,6, revealed a fragment with the unique lunar mineralogy of plagioclase, olivine, diopside, a Ti-Fe-Zr phase, and a trace of ilmenite. It should be emphasized that the ferromagnesian minerals have the most primitive composition in terms of the Mg/Fe ratio for any returned lunar sample with complex mineralogy, and that this fragment is a candidate for early material resulting from a lunar differentiation sequence. We report here the second occurrence of diopside and a third occurrence of a recently identified Ti-Fe-Zr phase, as well as the general character of this fragment.

The modal analysis of this fragment includes ~30% olivine, ~70% plagioclase, ~2% diopside, and about eight grains of an opaque phase (5 to 20 μ m). The texture is fine-grained (0.05 to 0.2 mm) and individual grains have a polygonal outline suggesting that recrystallization may have modified a previously existing texture.

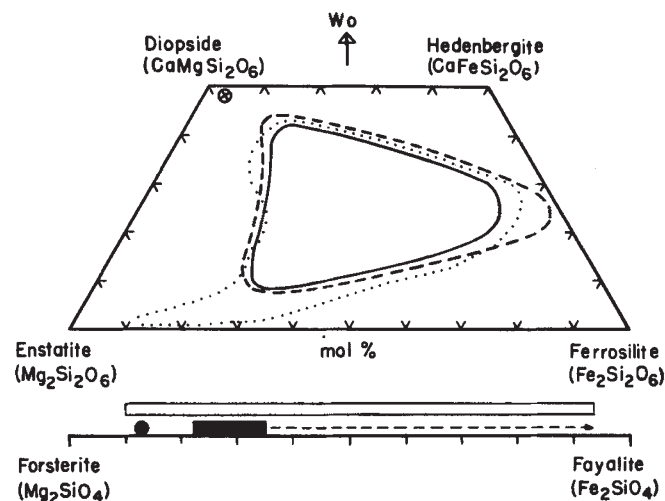


Fig. 1 Generalized pyroxene and olivine compositions for lunar material. Pyroxenes: ---, Apollo 11 basaltic rocks; —, Apollo 12 basaltic rocks; . . . , Apollo 14 basalts and breccias; ⊗, Diopside described in this report. Olivines: solid bar, early olivines in common lunar basalts; ---, late olivines in lunar basalts; open bar, olivines in lunar breccias; ●, olivine described in this communication.

Microprobe analyses of the four principal phases are given in Table 1. The olivine (analyses 1 and 2, Table 1) is comparable in Mg content (Fo 87) with the most Mg-rich lunar olivines (~Fo 90), and is more magnesian than olivines in common Apollo 11, 12 or 14 basaltic rocks (Fig. 1). Analyses 3 and 4 (Table 1) represent nearly pure diopside (En 48, Wo 48, Fs 4), a mineral only observed previously in one other lunar fragment⁴. X-ray scanning images of the diopside grains showed no intergrowths on the μ m scale. As for olivine, the high Mg content of the diopside compared with other lunar pyroxenes (Fig. 1) suggests a primitive origin. Four plagioclase analyses (10 to 13, Table 1) show a high Ca content (average An 96.1 mol. %), and a low minor element content (Mg ≈ 0.05 wt % Fe ≈ 0.09 wt %, K ≈ 0.11 wt %). These data are comparable with those (An, 96.7 to 99.0; Mg, 0.035; Fe, 0.1; K, 0.02) obtained from the Apollo 15 anorthosite⁵, 15415, and suggest that the origin of these two rock types may be related; the K content is, however, more similar to that in the high-K type of anorthosite⁶.

Table 1 Microprobe Analyses

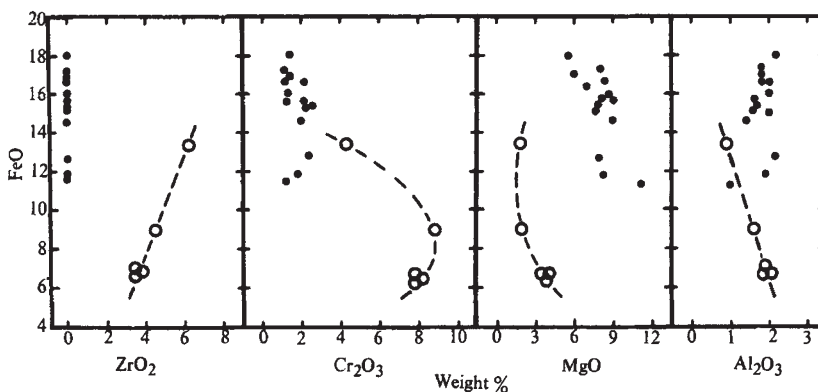
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Na ₂ O	0.0	0.0	0.13	0.15	0.04	0.05	0.0	—	—
MgO	46.9	47.1	17.6	17.2	3.74	3.76	3.57	1.7	1.9
Al ₂ O ₃	0.01	0.02	1.2	1.81	1.89	1.92	1.84	0.9	1.6
SiO ₂	41.7	41.5	54.1	53.7	0.22	0.17	0.18	0.2	0.6
CaO	0.16	0.14	23.9	23.9	3.36	3.07	3.02	3.1	3.1
TiO ₂	0.08	0.0	0.66	1.20	67.2	67.7	67.9	68.8	70.1
Cr ₂ O ₃	0.05	0.08	0.43	0.58	8.24	7.83	7.78	4.3	8.8
MnO	0.15	0.11	0.06	0.06	0.08	0.05	0.06	0.2	—
FeO	11.7	12.0	2.43	2.32	6.56	6.39	6.60	13.4	9.0
ZrO ₂	—	—	—	—	3.65	3.71	3.61	6.1	4.5
R.E.	—	—	—	—	—	—	—	1.3	—
Sum	100.7	100.9	100.5	100.9	95.0	94.6	94.6	100.0	99.6

	An(mol. %)	Ab(mol. %)	K(wt %)	Fe(wt %)	Mg(wt %)
(10)	96.5	4.5	0.11	0.11	0.11
(11)	96.0	4.5	0.11	0.09	0.05
(12)	96.0	4.5	0.10	0.08	0.01
(13)	96.0	5.0	0.10	0.06	0.04

(1) and (2), Olivine Fo 87; (3) and (4), diopside En 48 Wo 48 Fs 4; (5), (6) and (7), Ti-Fe-Zr phase; (8), analysis of phase X⁷ in 14310; (9), analysis of pseudo-armalcolite⁸ (calculated from molecular formula); (10) to (13), plagioclase.

The Ti-Fe-Zr mineral (analyses 5 to 7, Table 1) is compositionally similar to phase X⁷ and pseudo-armalcolite⁸ (analyses 8 and 9, respectively, Table 1) reported in other Apollo 14 samples. The three analyses determined in the present work have low oxide totals; a wavelength scan, however, showed the presence of about 1.0 wt % Ce₂O₃ and La₂O₃ and minor Pr, Nd and Sm which would account for

Fig. 2 ○, Principal element trends of Cr-Zr-armalcolite; ●, principal element trends of armalcolite.



some or all of this deficiency. In at least one grain, ilmenite was included within this Ti-Fe-Zr phase. It has been suggested⁸ that this phase is similar to armalcolite ($\text{Fe}_{0.5}\text{Mg}_{0.5}\text{Ti}_2\text{O}_5$), and provisionally it was named pseudo-armalcolite. The data for this phase converted to atom proportions based on 5(0) are given in Table 2. The Ti and Zr add up to about 2.0, and the remaining atoms to slightly less than 0.9, but these data may be modified by the presence of the rare earths as well as the possible presence of Ti^{3+} on the Moon⁹. Although the calculated formula is similar to armalcolite, the chemical trends of ZrO_2 , Cr_2O_3 , MgO and Al_2O_3 against FeO (Fig. 2) suggest complex chemical variations. The analyses of armalcolite¹⁰ did not reveal any ZrO_2 , whereas the new phase contains 4 to 6%. The MgO content is much higher in the armalcolite, and the Cr_2O_3 content much lower. Tentatively we use the name Cr-Zr-armalcolite for identification, but this

name has not been approved as that for a new mineral by the IMA Committee on Nomenclature. Two X-ray scanning images of the mineral assemblage (Ca K_α and Fe K_α) are shown in Figs. 3a and 3b while an identification key is shown in Fig. 3c.

Table 2 Atomic Proportions based on 5(0) for Cr-Zr-Armalcolite Analyses of Table 1

	(5)*	(6)*	(7)*	(8)*	(9)*
Ti	1.90	1.92	1.91	1.93	1.91
Zr	0.07	0.07	0.07	0.11	0.08
Mg	0.21	0.21	0.20	0.09	0.10
Al	0.09	0.09	0.08	0.04	0.07
Si	0.01	0.01	0.01	0.01	0.02
Ca	0.14	0.12	0.12	0.12	0.12
Cr	0.25	0.24	0.22	0.13	0.25
Mn	0.0	0.0	0.01	0.0	—
Fe	0.21	0.20	0.21	0.42	0.27

* See Table 1.

The relatively pure diopside analyses reported here represent the most Mg-rich calcic pyroxene reported from lunar material. This composition, together with those of the other principal phases, indicates that the low Fe/Mg ratio of this fragment is consistent with a product resulting from early crystal-liquid differentiation. The high chromium content of the Cr-Zr-armalcolite has been interpreted⁸ as an indication of a high formation temperature; this is consistent with the high temperature expected for the ferromagnesian minerals.

Tentatively we propose that the fragment was ejected during a deep impact (perhaps Imbrian) from an ultramafic region at depth in the Moon. The presence of such a layer was implied by the seismic studies of Toksöz *et al.*¹¹.

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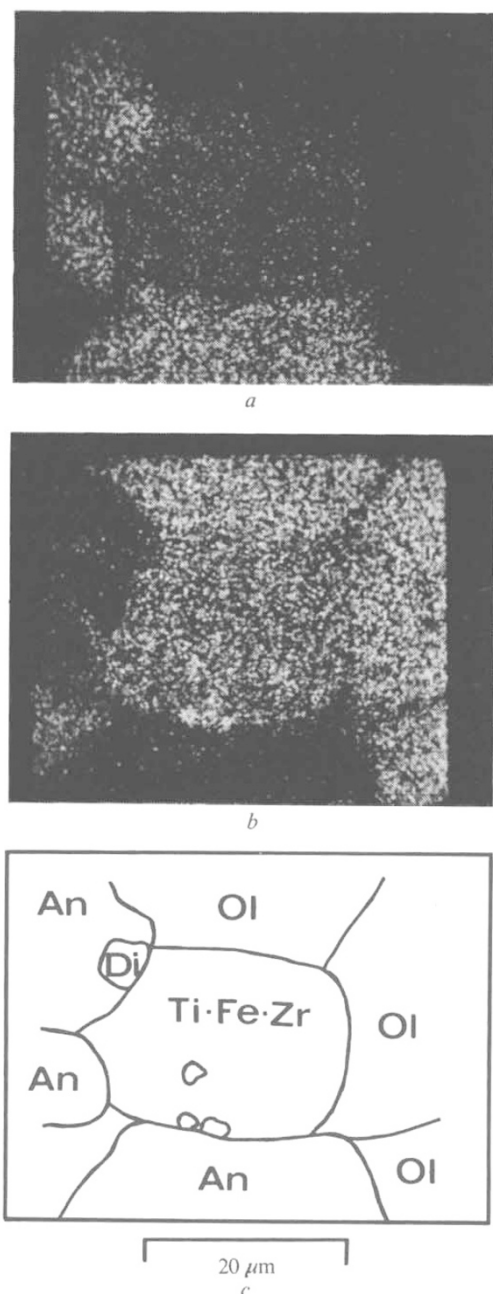


Fig. 3 a, Ca K_α scanning image of chief phases; b, Fe K_α scanning image of major phases; c, key for identification. An, Anorthite; Ol, olivine; Di, diopside; Ti-Fe-Zr, Cr-Zr-armalcolite. Small circles represent ilmenite in Ti-Fe-Zr.

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